Adapting Audio-JEPA for Low-Resource Thai Automatic Speech Recognition

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***Abstract*—Automatic speech recognition for Thai is limited by the scarcity of transcribed speech. Self-supervised pretraining reduces this dependence, but evidence for low-resource recognition comes mainly from contrastive and discrete-unit models. Audio-JEPA learns audio representations by predicting the latent features of masked spectrogram regions, but it was developed for general audio classification, and its coarse patches give too few time steps for character-level transcription. We adapt Audio-JEPA to speech by replacing its square spectrogram patches with a full-frequency projection stem that produces one token every 20 milliseconds, and by masking short temporal spans instead of two-dimensional blocks. The adapted encoder is pretrained on unlabeled Thai speech from the Porjai corpus, either from random initialization or from the original Audio-JEPA weights, and is then fine-tuned with a connectionist temporal classification head. We design a controlled comparison that separates the effects of upstream Audio-JEPA weights, in-domain Thai pretraining, and supervised training alone, using labeled budgets of 1, 10, and 50 hours and the full training set, evaluated with character and word error rates.**

I. Introduction

Automatic speech recognition (ASR) converts speech into text, but supervised systems typically depend on large amounts of transcribed audio. Accurate transcripts require time and human effort, so this requirement is difficult to meet when labeled speech is scarce [1]. Thai is such a case. Paired speech–text data remains limited compared with high-resource languages [2], and the language brings additional difficulties: it is tonal, distinguishes short and long vowels, and is written without spaces between words. In this work, "low-resource" refers primarily to a limited transcription budget. The availability of unlabeled audio and of computation is reported separately.

Conventional ASR describes the speech signal with fixed acoustic features such as mel-frequency cepstral coefficients (MFCCs) or Mel-spectrograms [3]. Self-supervised learning (SSL) instead learns representations directly from unlabeled audio, which is far cheaper to collect than transcripts. Models such as wav2vec 2.0 and HuBERT pretrain an encoder on untranscribed speech and then adapt it to recognition with few labeled examples [4], [5]. The question therefore shifts from which features to compute to how an encoder should learn its representations from unlabeled audio.

This paper follows a representation-based approach. Instead of reconstructing the input signal or predicting discrete pseudo-labels, the encoder learns by predicting the *representations* of hidden parts of the input from the visible context. Joint-embedding predictive architectures (JEPAs) implement this principle [6], [7], and Audio-JEPA applies it to audio by predicting latent features of masked Mel-spectrogram patches [8]. Audio-JEPA learns competitive general-purpose audio representations with a modest training budget. However, it was pretrained on general audio rather than speech, evaluated mainly on classification tasks, and never applied to continuous transcription or to Thai. Its coarse patch grid also produces one time step every 625 ms, which is too coarse for character-level CTC alignment. Whether JEPA-style representation prediction can reduce the labeled-data requirement of Thai ASR is therefore an open question.

We investigate this question with a two-stage framework. First, a speech-adapted Audio-JEPA is pretrained, or further pretrained, on unlabeled Thai speech. Second, its encoder is transferred to a recognizer with a connectionist temporal classification (CTC) head [9] and fine-tuned under controlled labeled-data budgets. The contributions of this work are as follows:

- We adapt Audio-JEPA to speech by replacing its 16×16 spectrogram patches with a full-frequency projection stem that produces one token every 20 ms, and by replacing two-dimensional block masking with short temporal-span masking. All Transformer blocks of the original model can still be transferred.
- We define a JEPA pretraining and CTC fine-tuning pipeline for Thai, including valid-length handling, CTC feasibility checks, and representation-collapse monitoring.
- We design a controlled comparison that separates the contribution of upstream Audio-JEPA weights, in-domain Thai JEPA pretraining, and supervised training alone, evaluated with 1, 10, and 50 hours and the full set of labeled Thai speech using character and word error rates.

The rest of the paper is organized as follows. Section II reviews related work. Section III describes the proposed method and experimental design. Section IV will report the experimental results, and Section V will conclude the paper.

II. Related Work

A. Speech Representation Learning

Early ASR systems extracted fixed features such as MFCCs [3] and PLP [10] and modeled them with HMM–GMM

systems [11], [12], which kept feature extraction, acoustic modeling, and language modeling separate. Supervised neural models replaced these components: deep, convolutional, and recurrent networks improved acoustic modeling [13]–[15], CTC removed the need for frame-level alignments [9], and attention-based and Conformer models further simplified and strengthened recognition [16], [17]. All of these models, however, require large amounts of transcribed speech. Self-supervised learning (SSL) reduces this requirement by deriving training targets from unlabeled audio [18].

B. Self-Supervised Learning Paradigms

SSL methods for speech differ mainly in what they predict. *Reconstruction* methods such as Mockingjay, DeCoAR, and TERA restore masked or corrupted input frames [19]–[21]. They are simple, but they spend capacity on noise and low-level acoustic detail. *Contrastive* methods such as CPC and wav2vec 2.0 select the true latent target from distractors [4], [22], which makes them sensitive to negative sampling and quantization choices. *Discrete-unit* methods such as HuBERT and WavLM predict cluster assignments at masked positions [5], [23], so their quality depends on the offline clustering. *Representation-prediction* methods such as data2vec and JEPA regress continuous features produced by an EMA target encoder [6], [7], [24]. They need no reconstruction, negatives, or clustering, and this is the paradigm we adopt.

C. JEPA and Audio-JEPA

I-JEPA predicts the embeddings of masked image blocks from a visible context block [7]. It avoids both the augmentation biases of invariance-based methods and the pixel-level burden of generative methods. A-JEPA and the study of Riou et al. carried this idea to audio spectrograms and showed that the choice of masking and target regions strongly affects representation quality [25], [26]. Audio-JEPA treats a Mel-spectrogram as an image of 16×16 patches and is pretrained on about 5,300 hours of AudioSet [8]. It is competitive with wav2vec 2.0 and data2vec on several X-ARES tasks, but weaker on fine-grained speech tasks, and it has been evaluated only on classification-style tasks. Speech-specific JEPAs that add GMM-based soft targets report ASR gains [27], [28], but they rely on tens of thousands of hours of English speech.

D. SSL for Low-Resource and Thai ASR

Pretraining on unlabeled audio followed by CTC fine-tuning on a small labeled set is the standard recipe for low-resource ASR. Wav2vec 2.0 reached 4.8%/8.2% WER on LibriSpeech test-clean/test-other with only ten minutes of labels, using 53,000 hours of unlabeled audio and a language model [4], and HuBERT matched or exceeded it across label budgets [5]. Multilingual pretraining (XLSR, XLS-R, MMS) extends these gains to languages with little data [1], [29], [30], whereas Whisper relies on large-scale weak supervision [31]. For Thai, XLSR-Thai continues pretraining on 36,000 hours of Thai speech [2], and Typhoon ASR shows the importance of consistent Thai text normalization [32]. Public corpora such as Porjai [33] now make controlled label-efficiency studies possible.

E. Research Gap

Evidence that SSL improves low-resource ASR comes almost entirely from contrastive and discrete-unit models. Audio-JEPA learns audio representations efficiently, but it was trained on general audio, evaluated on classification, uses a temporal resolution too coarse for CTC, and has not been applied to Thai. To our knowledge, JEPA-style representation prediction has not been evaluated for low-resource Thai ASR.

III. Methodology

A. Overview

We adopt the representation-based approach introduced in Section I: an encoder first learns speech representations from unlabeled Thai audio, and the same encoder is then adapted to transcription using a limited amount of labeled speech. Fig. 1 summarizes the two stages. In Stage 1 (pretraining), log-Mel features computed from untranscribed Thai recordings are passed to a speech-adapted Audio-JEPA, which learns by predicting the latent features of hidden time spans [8]. In Stage 2 (fine-tuning), the pretrained context encoder is transferred to a recognizer, a connectionist temporal classification (CTC) head [9] is attached, and the model is trained on Thai transcripts under fixed labeled-data budgets. The JEPA predictor and target encoder are discarded after Stage 1, so they add no inference cost.

B. Dataset and Text Processing

Experiments use the Porjai Central Thai speech corpus released by CMKL [33]. Because the corpus does not provide speaker identifiers, utterances are assigned to training, validation, and test partitions (80%/10%/10%) by a deterministic hash of the utterance identifier. Every run therefore uses identical partitions. A consequence is that the partitions are utterance-disjoint but not guaranteed to be speaker-disjoint; we treat this as a limitation when interpreting test results. Self-supervised pretraining reads audio only from the training partition. Validation and test audio is never seen during Stage 1, and transcripts are never seen in Stage 1 at all.

Audio is decoded, converted to mono, and resampled to 16 kHz. Utterances outside the permitted duration range are removed from training instead of being truncated, so every training waveform remains consistent with its full transcript. The number of excluded utterances and hours is logged for each run.

Transcripts are normalized with a single rule set that is applied identically to training labels, references, and hypotheses: Unicode NFKC normalization, removal of zero-width characters and punctuation, mapping of Thai digits to Arabic digits, lowercasing of Latin letters, and collapsing of repeated whitespace. Thai vowels, tone marks, and the repetition mark are retained. The output vocabulary is character-level and includes Thai consonants, vowels, tone and orthographic marks, digits, Latin letters, a word-boundary space token, an unknown

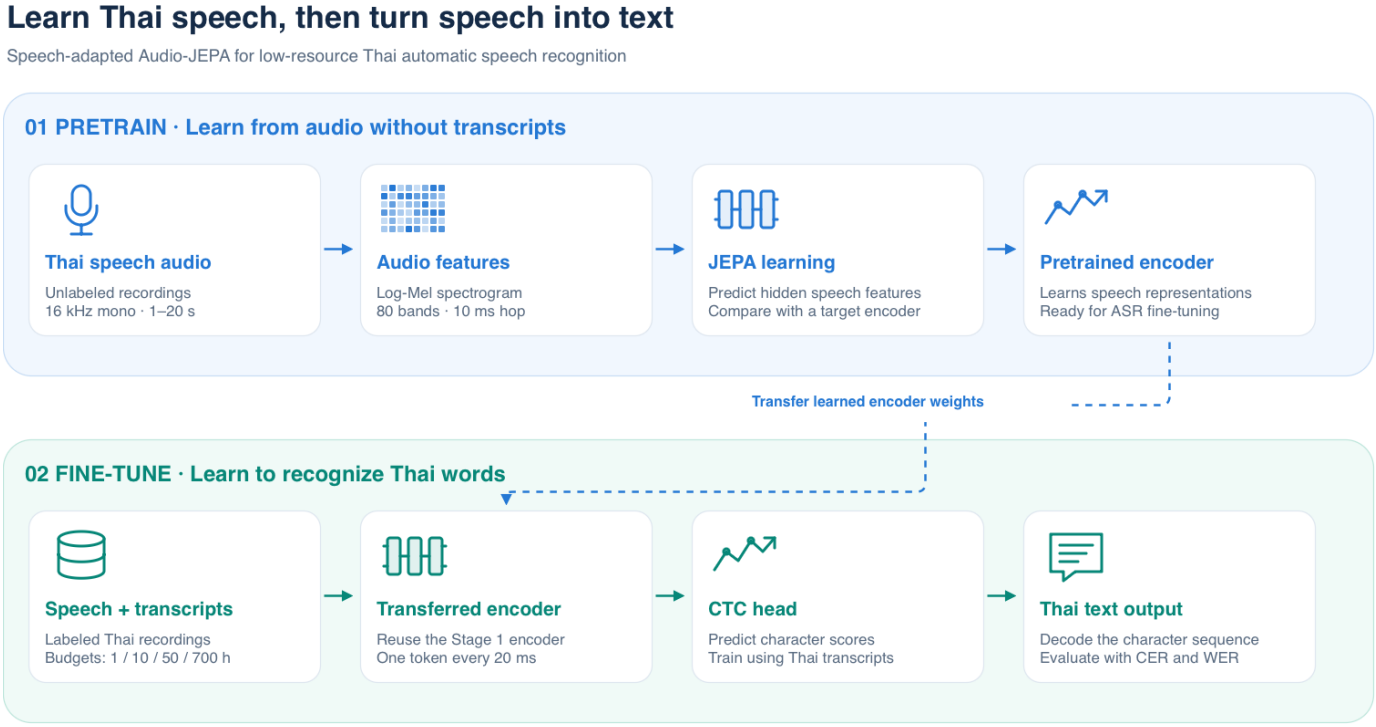


Fig. 1. Overall framework. Stage 1 pretrains a speech-adapted Audio-JEPA encoder on unlabeled Thai audio. Stage 2 transfers the learned encoder weights to a CTC recognizer trained with 1, 10, 50, or approximately 700 hours of transcribed Thai speech and evaluated with CER and WER.

token, and the CTC blank at index 0. Characters outside the vocabulary map to the unknown token during training, but references keep their original characters, so such characters still count as errors at evaluation.

C. Acoustic Frontend and Speech Tokenization

Each waveform is converted to 80 log-Mel filterbank features using a 25 ms analysis window and a 10 ms hop. The original Audio-JEPA divides a 256×128 spectrogram into 16×16 patches, which yields only 16 temporal positions for a 10-second clip, or one position every 625 ms [8]. CTC requires that the number of output frames T satisfies

$$T \geq U + R, \tag{1}$$

where U is the transcript length and R is the number of adjacent repeated labels, because repeated labels must be separated by a blank. A character-level Thai transcript typically has more characters than the coarse grid has time steps, so most utterances cannot be aligned at that resolution.

We therefore replace the square patch embedding with a full-frequency temporal projection stem. A two-dimensional convolution with kernel and stride $(80, 2)$ and 768 output channels maps each pair of consecutive 10 ms frames, across all 80 Mel bands, to one 768-dimensional token. The encoder thus receives one token every 20 ms (50 tokens per second), as shown in Fig. 2. The exact number of valid tokens is computed from the unpadded waveform length:

$$F = \left\lfloor \frac{N - W}{H} \right\rfloor + 1, \qquad T = \left\lfloor \frac{F - 2}{2} \right\rfloor + 1, \tag{2}$$

where N is the number of samples, W and H are the window and hop lengths in samples, F is the number of filterbank frames, and T is the number of encoder tokens. These valid lengths, rather than padded tensor lengths, define the attention masks, the masking region in Stage 1, and the CTC input lengths in Stage 2. Before training, every batch is checked against (1), and the ratio $T/(U + R)$ is reported.

D. Encoder Architecture

The encoder keeps the Audio-JEPA Vision Transformer backbone: 12 pre-LayerNorm Transformer blocks with width 768, 12 attention heads, and a feed-forward width of 3072 with GELU activations, followed by a final LayerNorm [8]. Fixed sinusoidal positional embeddings are added to the tokens along the time axis. Padded positions are excluded from self-attention by an additive attention mask. The encoder has no classification token and performs no temporal pooling, so its output is an ordered sequence of 20 ms frame representations. The model attends bidirectionally and targets offline transcription.

E. Stage 1: Thai Speech JEPA Pretraining

Fig. 2 details the pretraining objective. Let P_θ denote the projection stem and E_θ the context encoder. A target copy $(P_{\bar{\theta}}, E_{\bar{\theta}})$ is initialized with identical weights and receives no gradients.

1) Short-span temporal masking: The original Audio-JEPA samples random two-dimensional blocks on the spectrogram patch grid [8]. Speech information is organized mainly along

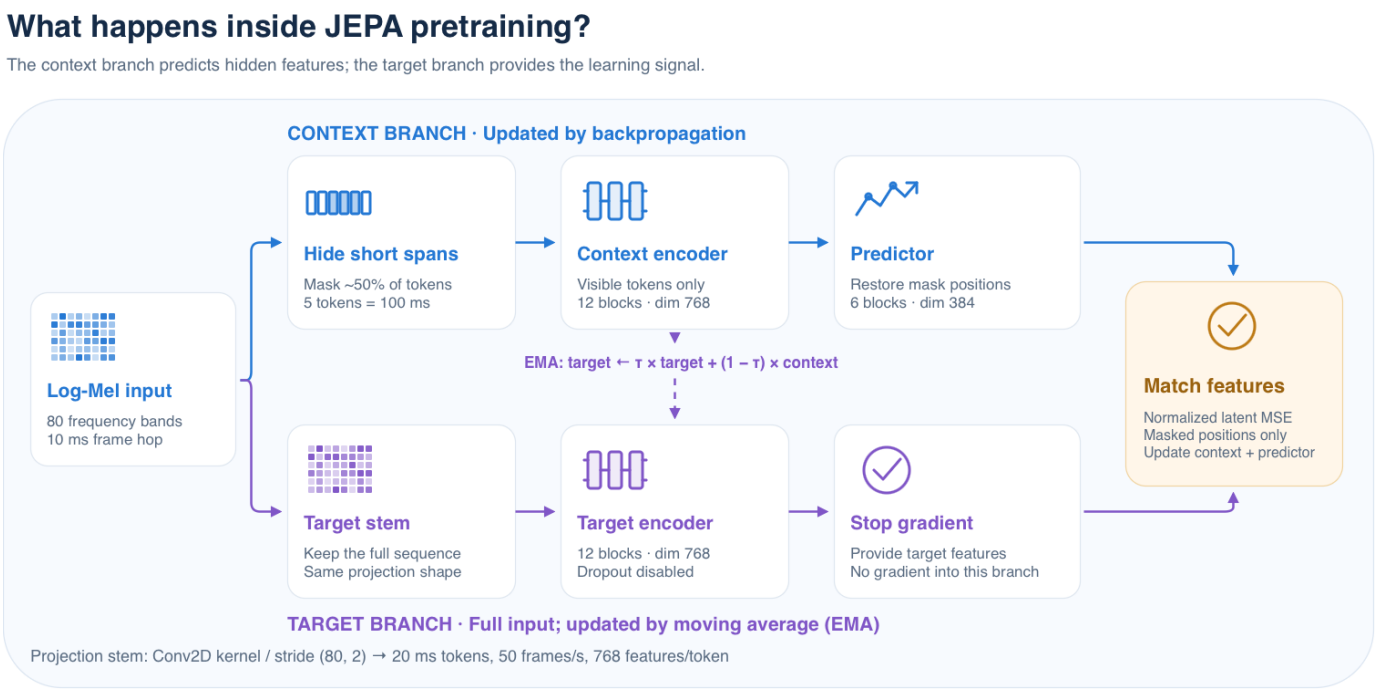


Fig. 2. Inside JEPA pretraining. The context branch sees only visible tokens and predicts latent features at the hidden positions. The target branch encodes the complete sequence and provides stop-gradient targets. Gradients update the context stem, context encoder, and predictor. The target branch is updated only by an exponential moving average (EMA) of the context weights.

time, so we mask whole time steps across all frequencies. For each batch, spans of five consecutive tokens (100 ms, roughly phone-to-syllable scale) are sampled within the valid region until approximately 50% of the valid tokens are hidden. The context branch receives only the remaining visible tokens, each with its original positional embedding. Prediction targets are drawn from the hidden spans, with two target sets per batch. Padding is never selected as context or target. Hiding short spans prevents the model from solving the task by copying immediately adjacent frames, while leaving enough surrounding context to infer the hidden content.

2) Predictor: The predictor is a narrower Transformer with 6 blocks of width 384 and 12 heads. Context outputs are linearly projected to width 384. A shared learnable mask token, plus the positional embedding of each hidden position, is appended for every target position, and the sequence is processed by the predictor blocks. A final linear layer maps the outputs back to 768 dimensions, giving the prediction $\hat{z}_t$ for each target position t.

3) Latent prediction objective: Let z_t be the target encoder output at position t, and let $\bar{z}_t$ be z_t standardized across its channels. Following the normalized latent loss of Audio-JEPA, we minimize the squared distance between ℓ_2-normalized prediction and target vectors over the set of target positions M:

$$\begin{aligned}\mathcal{L}_{\text{JEPA}} &= \frac{1}{|M|}\sum_{t\in M}\left\|\frac{\hat{z}_t}{\|\hat{z}_t\|_2} - \text{sg}\left[\frac{\bar{z}_t}{\|\bar{z}_t\|_2}\right]\right\|_2^2 \\ &= \frac{1}{|M|}\sum_{t\in M}\big(2-2\cos(\hat{z}_t,\bar{z}_t)\big),\end{aligned} \tag{3}$$

where $\text{sg}[\cdot]$ stops gradients. The loss is computed only at hidden positions and is averaged over the target sets. Because the targets are features rather than spectrogram values, the model is not required to reproduce low-level acoustic detail such as background noise.

4) EMA target update: Only the context stem, context encoder, and predictor are optimized. After each update, the target parameters follow an exponential moving average of the context parameters:

$$\bar{\theta} \leftarrow \tau_s\bar{\theta} + (1-\tau_s)\theta, \qquad \tau_s = 1-(1-\tau_0)\,\frac{1+\cos(\pi s/S)}{2}, \tag{4}$$

with $\tau_0 = 0.996$, so τ_s rises from 0.996 to 1.0 over the S pretraining updates. Dropout is disabled in the target branch.

5) Collapse monitoring: Latent prediction objectives admit a trivial solution in which all embeddings become constant. The EMA target and stop-gradient discourage this solution but do not rule it out. During pretraining we therefore log the per-channel variance of target features, the mean cosine similarity between utterance-level embeddings within a batch, and the mean feature norm. A run whose loss falls while feature variance approaches zero or inter-utterance similarity approaches one is treated as collapsed and is not used for fine-tuning.

6) Initialization: Stage 1 is run from two initializations. In the transfer setting, all Transformer blocks and the final LayerNorm are loaded from the public Audio-JEPA checkpoint pretrained on AudioSet [8], [34]. The projection stem cannot be loaded because its kernel shape differs from the original

16×16 patch embedding, so it is initialized randomly. The pretraining then continues on Thai audio. In the scratch setting, all weights are initialized randomly. A loading report records every loaded, reinitialized, and skipped tensor together with the fraction of parameters transferred.

F. Stage 2: CTC Fine-Tuning

After pretraining, the context stem and context encoder are kept. The predictor and target branch are removed. A dropout layer ($p = 0.1$) and a linear layer map each 768-dimensional frame representation to a distribution over the vocabulary $\mathcal{V}$ and the blank symbol $\varnothing$. For an input x with transcript y, training minimizes the CTC loss

$$\mathcal{L}_{\mathrm{CTC}} = -\log \sum_{\pi \in \mathcal{B}^{-1}(y)} \prod_{t=1}^{T} p(\pi_t \mid x), \tag{5}$$

where $\mathcal{B}$ merges consecutive identical labels and then removes blanks [9]. Input lengths are the valid token counts defined in Section III-C. Infinite losses are not replaced with zero, so any alignment failure is visible in the logs.

Because the projection stem has been retrained or newly initialized, the optimizer uses three parameter groups with separate learning rates: the Transformer backbone, the projection stem, and the CTC head. Pretrained backbones use a learning rate one-tenth of the head learning rate, whereas the stem and head use larger rates. When a frozen-encoder variant is run, only the Transformer blocks are frozen, and the stem and head remain trainable. Inference uses greedy CTC decoding: the most probable label is selected at each valid frame, repeats are merged, and blanks are removed. No external language model is used.

G. Experimental Configurations

Table I lists the systems. All systems share the same frontend, partitions, vocabulary, CTC head, and decoding procedure. Comparing them isolates three factors: the value of upstream Audio-JEPA weights (A versus B), the value of in-domain Thai JEPA pretraining (B versus C), and whether Thai JEPA pretraining alone, without AudioSet weights, is sufficient (A versus D). A wav2vec 2.0 encoder fine-tuned with an identical CTC head serves as a speech-specialized self-supervised reference [4]. Its pretraining data and model size differ from those of our encoder, so it is a practical reference point rather than a controlled comparison of objectives.

To measure label efficiency, CTC fine-tuning is repeated with nested labeled subsets of 1, 10, and 50 hours and with the full training partition of approximately 700 hours. Each smaller subset is contained in every larger one. Self-supervised pretraining always uses the same unlabeled training audio, so only the transcription budget varies across conditions. Each configuration is trained with three random seeds (42, 43, and 44), and results are reported as mean $\pm$ standard deviation.

TABLE I
EXPERIMENTAL CONFIGURATIONS. ALL SYSTEMS END WITH A NEWLY INITIALIZED CTC HEAD.

ID	Encoder initialization	Purpose
A	Random	Supervised-only control
B	Audio-JEPA (AudioSet)	Effect of upstream transfer
C	Audio-JEPA $\rightarrow$ Thai JEPA	Effect of in-domain Thai pretraining (proposed)
D	Random $\rightarrow$ Thai JEPA	Thai pretraining without AudioSet weights
Ref	wav2vec 2.0	Speech-specialized SSL reference

TABLE II
OPTIMIZATION SETTINGS FOR BOTH TRAINING STAGES.

Setting	Value
Stage 1: JEPA pretraining	
Optimizer	AdamW, learning rate 1×10^{-4}, weight decay 0.01
Batch / epochs	8 utterances per batch; 8 epochs
Masking	5-token (100 ms) spans, about 50% of valid tokens; 2 target sets
Predictor	6 blocks, width 384, 12 heads
EMA	τ: 0.996 $\rightarrow$ 1.0, cosine schedule
Stage 2: CTC fine-tuning	
Learning rates	Head 3×10^{-4}; stem 3×10^{-4} (1×10^{-4} after Thai JEPA); backbone 3×10^{-5} if pretrained, 3×10^{-4} if random
Batch / epochs	8 utterances, up to 15 epochs (1 h budget: 4 utterances, up to 25 epochs)
Early stopping	Patience of 5 validation evaluations (8 for the 1 h budget)
Regularization	Dropout 0.1 before the CTC head; weight decay 0.01
Decoding	Greedy CTC, no language model

H. Training Details

Table II lists the optimization settings, taken from the experiment configurations in our codebase. Both stages use AdamW with a linear warmup over the first 10% of updates, followed by cosine decay to 1% of the peak learning rate, and gradient clipping at norm 1.0. Weight decay is not applied to biases or normalization parameters. The checkpoint with the lowest validation CER is selected, and training stops early when validation CER stops improving.

I. Evaluation Metrics

The primary metric is the character error rate (CER). Because Thai is written without spaces between words, word error rate (WER) requires word segmentation. We segment both references and hypotheses with the PyThaiNLP `newmm` dictionary-based tokenizer [35], with the same version and dictionary for all systems, and then compute WER. For both units,

$$\mathrm{ER} = \frac{S + D + I}{N} \times 100\%, \tag{6}$$

where S, D, and I are the numbers of substitutions, deletions, and insertions, and N is the number of reference units. Counts are accumulated over the whole evaluation set rather than averaged per utterance. Model selection uses only the

validation partition. The test partition is evaluated once, with the selected checkpoint of each run.

IV. Experimental Results and Discussion

This section will be completed after the experiments have been conducted.

V. Conclusion and Future Work

This section will be completed after the experiments have been conducted.

Acknowledgment

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

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